

Solutions to MATH5003P

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Abstract

This is a collection of personal solutions to homework assigned in MATH5003P, a graduate-level introduction to differential manifolds at USTC.

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1 HOMEWORK FOR LECTURE 1

Exercise 1.1. Prove that, for $1 \leq k \leq n$, the Grassmannian

$$\text{Gr}_{\mathbb{R}}(k, n) = \{k\text{-dimensional linear subspace of } \mathbb{R}^n\}$$

is a smooth manifold, by explicitly constructing open cover and local charts.

Solution. We first endow $\text{Gr}_{\mathbb{R}}(k, n)$ with a topology. Note that k -frames in $\mathbb{R}^n$ form an open subset X of matrix space $\mathbb{R}^{nk}$ (actually, X is called the Stiefel manifold). A k -dimensional subspace of $\mathbb{R}^n$ is determined by a k -frame up to a left action by $\text{GL}(k, \mathbb{R})$. As a result, we identify $\text{Gr}_{\mathbb{R}}(k, n)$ with the quotient $q: X \rightarrow X/\text{GL}(k, \mathbb{R})$. We check that $\text{Gr}_{\mathbb{R}}(k, n)$ is Hausdorff. It suffices to show that $\Delta \subset \text{Gr}_{\mathbb{R}}(k, n) \times \text{Gr}_{\mathbb{R}}(k, n)$ is closed. Indeed, $(q \times q)^{-1}(\Delta) \subset X \times X$ consists of pairs of k -frames (f_1, f_2) that span the same subspace. The closedness is checked by taking limits.

Given a subspace $V_0 \leq \mathbb{R}^n$, we take its orthogonal decomposition $V_0^\perp$ with respect to the standard inner product. Let $p: \mathbb{R}^n \rightarrow V$ be the orthogonal projection. Consider

$$\varphi: \text{Hom}(V_0, V_0^\perp) \rightarrow \text{Gr}_{\mathbb{R}}(k, n), L \mapsto \{(v, Lv) \mid v \in V_0\}.$$

Denote its image by U . $q^{-1}(U)$ consists of k -frames f such that $p(f)$ spanning V_0 . Conversely, as each $p_V: V \rightarrow V_0$ is an isomorphism for $V \in U$, the inverse map is given by $\phi: V \mapsto L: v \mapsto p_V^{-1}(v) - v$.

ϕ is continuous. Indeed, φ can be lifted to $\tilde{\phi}: q^{-1}(U) \rightarrow \text{Hom}(V_0, V_0^\perp)$, since a linear map is determined by its image on a basis. On the other hand, fix a k -frame $(v_1, 0), \dots, (v_k, 0)$ in $q^{-1}(V)$. Let $\tilde{\varphi}$ sending $L \in \text{Hom}(V_0, V_0^\perp)$ to $(v_1, Lv_1), \dots, (v_k, Lv_k)$. Then $\tilde{\varphi}$ is continuous and $\varphi = q \circ \tilde{\varphi}$.

It remains to verify the transition maps. Let $\phi: U \rightarrow \text{Hom}(V_0, V_0^\perp), \phi': U' \rightarrow \text{Hom}(V'_0, (V'_0)^\perp)$ be two local charts as constructed before. We can take $T \in \text{GL}(n, \mathbb{R})$ sending V_0 to V'_0 . As a result, $TLv = L'Tv$ and $L' = TLT^{-1}$, which implies the smoothness of transition map. $\square$

Exercise 1.2. Let M be a smooth manifold and $\phi \in \text{Diff}(M)$. Prove that its graph $\text{graph}(\phi) = \{(x, \phi(x)) \mid x \in M\}$ is a smooth manifold.

Solution. Regard $\text{graph}(\phi)$ as a subspace of $M \times M$. Then $s: M \rightarrow \text{graph}(\phi), x \mapsto (x, \phi(x))$ and $\pi: \text{graph}(\phi) \rightarrow M, (x, y) \mapsto x$ give a homeomorphism, which induces a manifold structure on $\text{graph}(\phi)$. $\square$

Exercise 1.3. Let M be a closed smooth manifold and $\phi \in \text{Diff}(M)$. Prove that the mapping torus defined by

$$T_\phi(M) = [0, 1] \times M / \sim$$

is a smooth manifold, where $(0, x)$ is identified with $(1, \phi(x))$ for any $x \in M$.

Solution. Let $\pi: [0, 1] \times M \rightarrow T_\phi(M)$ be the quotient map and take $[(t, x)] \in T_\phi(M)$. First assume $t \in (0, 1)$. Then we give $(0, 1) \times M$ the structure of product manifold. Now consider $[(0, x)]$. Let $\varphi: U \rightarrow \mathbb{R}^n$ be a local chart near x . Then $\varphi \circ \phi^{-1}$ is a local chart near $\phi(x)$. Let $V = \{[(t, y)] \mid 0 \leq t < \frac{1}{2}, y \in U\} \cup \{[(t, y)] \mid \frac{1}{2} < t \leq 1, y \in \phi(U)\}$ and consider $\Phi: V \rightarrow (-\frac{1}{2}, \frac{1}{2}) \times \mathbb{R}^n$, defined by

$$\Phi([(t, y)]) = \begin{cases} (t, \varphi(x)), 0 \leq t < \frac{1}{2} \\ (t - 1, \varphi \circ \phi^{-1}(x)), \frac{1}{2} < t \leq 1. \end{cases}$$

As $\pi^{-1}(V) = [0, \frac{1}{2}) \times U \sqcup (\frac{1}{2}, 1] \times \phi(U)$, Φ can be lifted to a continuous map, so Φ is also continuous. One can check that Φ is a homeomorphism, with inverse given by

$$(s, p) \mapsto \begin{cases} [(s, \varphi^{-1}(p))], 0 \leq s < \frac{1}{2} \\ [(s + 1, \phi \circ \varphi^{-1}(p))], -\frac{1}{2} < s \leq 0. \end{cases}$$

It suffices to check the smoothness of transition maps we constructed above. Let $\Phi: V \rightarrow (-\frac{1}{2}, \frac{1}{2}) \times \mathbb{R}^n$, $\Psi: W \rightarrow (-\frac{1}{2}, \frac{1}{2}) \times \mathbb{R}^n$ be local charts. Then

$$\psi^{-1} \circ \Phi(s, p) \mapsto (s, \psi^{-1} \circ \varphi(p))$$

is a diffeomorphism. □

Exercise 1.4. Prove that the following set of matrices

$$H = \left\{ \begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix} \mid x, y, z \in \mathbb{R} \right\}$$

is a Lie group. H is usually called the Heisenberg group.

Solution. H admits the group structure under the operation

$$\begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & x' & z' \\ 0 & 1 & y' \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & x + x' & xy' + z + z' \\ 0 & 1 & y + y' \\ 0 & 0 & 1 \end{pmatrix}$$

The inverse map writes

$$\begin{pmatrix} 1 & -x & xy - z \\ 0 & 1 & -y \\ 0 & 0 & 1 \end{pmatrix}.$$

The two maps are smooth. $\square$

Exercise 1.5. Assume the orthogonal group $O(n) = \{A \in M(n, \mathbb{R}) \mid AA^T = I_n\}$ is a compact Lie group of dimension $\frac{1}{2}n(n-1)$. Prove that the special orthogonal group

$$SO(n) = \{A \in O(n) \mid \det A = 1\}$$

is a compact Lie group and calculate its dimension.

. $SO(n)$ is a connected component of $O(n)$ (this can be seen using canonical form of orthogonal matrices). As a consequence, $SO(n)$ is a closed subset of $O(n)$ and then is compact. Meanwhile, $SO(n)$ is an open subset of $O(n)$ and also a subgroup of $O(n)$, so $SO(n)$ is a compact Lie group with $\dim SO(n) = \frac{1}{2}n(n-1)$. $\square$

Exercise 1.6. Prove that $SO(3)$ is diffeomorphic to $\mathbb{RP}^3$ as smooth manifolds.

Solution. Recall that $SU(2)$ is diffeomorphic to S^3 . As $SU(2)$ is a double cover of $SO(3)$ and S^3 is a double cover of $\mathbb{RP}^3$, $SO(3)$ is diffeomorphic to $\mathbb{RP}^3$ as smooth manifolds.

A more concrete approach is to identify $SO(3)$ with $S^2 \times [0, 2\pi] / \sim$, where $(v, \theta) \sim (-v, 2\pi - \theta)$. Let $\varphi: SO(3) \rightarrow \mathbb{RP}^3, [(v, \theta)] \mapsto [(v \sin \frac{\theta}{2}, \cos \frac{\theta}{2})]$. $\square$

Exercise 1.7. Identify $\mathbb{CP}^n$ with $(\mathbb{C}^{n+1} \setminus \{0\}) / \mathbb{C}^\times$. Consider the map $S: \mathbb{CP}^1 \times \mathbb{CP}^1 \rightarrow \mathbb{CP}^3$ by

$$([(w_0, w_1)], [z_0, z_1]) \mapsto [(w_0 z_0, w_0 z_1, w_1 z_0, w_1 z_1)].$$

Prove that S is a smooth map. S is actually injective and is usually called the Segre embedding.

Solution. Assume that $w_1, z_1 \neq 0$. Then $w_1 z_1 \neq 0$, so near $[(w_0, w_1)], [z_0, z_1]$, the map is written as

$$(x, y) \mapsto (xy, x, y),$$

which is smooth. $\square$

Exercise 1.8. Consider group $E(n) = \mathbb{R}^n \rtimes \mathrm{O}(n)$ where the multiplication is given by

$$(v, A) \cdot (w, B) = (v + Aw, AB).$$

Check that $E(n)$ is a Lie group. Meanwhile, a representation of $E(n)$ is a Lie group homomorphism from $E(n)$ to $\mathrm{GL}(k, \mathbb{R})$ for some $k > 0$. Construct a representation of $E(n)$ that is injective.

Solution. As a manifold, $E(n) \cong \mathbb{R}^n \times \mathrm{O}(n)$. $E(n)$ is indeed a group with identity $(0, I_n)$. The group operation is smooth, and $(v, A)^{-1} = (-A^{-1}v, A^{-1})$.

Consider

$$\begin{aligned} \rho: E(n) &\rightarrow \mathrm{GL}(n+1, \mathbb{R}), \\ (v, A) &\mapsto \begin{pmatrix} A & v \\ 0 & 1 \end{pmatrix}. \end{aligned}$$

ρ is an injective representation of $E(n)$. □

Exercise 1.9. Prove that the upper half-plane in $\mathbb{C}$, defined by

$$\mathbb{H} = \{z \in \mathbb{C} \mid \mathrm{Im}(z) > 0\}$$

is a homogeneous space.

Solution. Consider $G = \mathrm{PGL}(2, \mathbb{R})$, which is the group of Möbius transformations on $\mathbb{H}$. The action is smooth and for every $z = x + \sqrt{-1}y \in \mathbb{H}$, the transformation given by

$$\begin{pmatrix} \sqrt{y} & \frac{x}{\sqrt{y}} \\ 0 & \frac{1}{\sqrt{y}} \end{pmatrix}$$

maps $\sqrt{-1}$ to z . Since z is arbitrary, we see that G acts transitively on $\mathbb{H}$. □

Exercise 1.10. Prove that if M and N are diffeomorphic, then $\dim M = \dim N$.

Solution. Let $f: M \rightarrow N$ be a diffeomorphism and $p \in M$. $df_p: T_p M \rightarrow T_{f(p)} N$ is an isomorphism, so $\dim M = \dim T_p M = \dim T_{f(p)} N = \dim N$. □

2 HOMEWORK FOR LECTURE 2

Exercise 2.1. Given the Grassmannian $\text{Gr}_{\mathbb{R}}(k, n)$, consider the following set

$$\gamma_{\mathbb{R}}(k, n) = \{(V, v) \in \text{Gr}_{\mathbb{R}}(k, n) \times \mathbb{R}^n \mid v \in V\}.$$

Prove that under the natural projection $\pi(V, v) = V$, $\gamma_{\mathbb{R}}(k, n)$ is a real vector bundle of rank k over $\text{Gr}_{\mathbb{R}}(k, n)$. This vector bundle is called the tautological bundle over $\text{Gr}_{\mathbb{R}}(k, n)$.

$\text{Gr}_{\mathbb{R}}(k, n)$ admits a local chart near V_0 , namely, $\varphi: U \rightarrow \text{Hom}(V_0, V_0^\perp)$, where $U = \{V \in \text{Gr}_{\mathbb{R}}(k, n) \mid V \cap V_0^\perp = 0\}$. Let $p: \mathbb{R}^n \rightarrow V_0$ be the orthogonal projection. Then $\Phi: \pi^{-1}(U) \rightarrow U \times V_0$, $(V, v) \mapsto (V, pv)$ gives a local trivialization.

Let $\varphi': U' \rightarrow \text{Hom}(V'_0, (V'_0)^\perp)$ be another chart with $U \cap U' \neq \emptyset$. Then the transition function is given by $\Phi' \circ \Phi^{-1}(v) = p' \circ p^{-1}(v)$, where $p: \mathbb{R}^n \rightarrow V'_0$ is the orthogonal projection. $\square$

Exercise 2.2. Let X, Y be vector fields on M , and locally (within some (U_α, ϕ_α)) write $X = (X_1, \dots, X_n)$ and $Y = (Y_1, \dots, Y_n)$ where X_i, Y_j are smooth functions on U_α . Prove that the Lie bracket locally writes as follows,

$$[X, Y] = (D_X Y_1 - D_Y X_1, \dots, D_X Y_n - D_Y X_n).$$

Use this to calculate $[X, Y]$ for $X, Y \in \Gamma(T\mathbb{R}^3)$ where

$$X(x, y, z) = (-y, x, 0), Y(x, y, z) = (0, -z, y).$$

Solution. Let f be a smooth function defined on some neighborhood of U .

$$\begin{aligned} [X, Y]f &= XYf - YXf = X \sum_{i=1}^n Y_i \frac{\partial f}{\partial u_i} - Y \sum_{i=1}^n X_i \frac{\partial f}{\partial u_i} \\ &= \sum_{i,j} \left(X_i Y_j \frac{\partial^2 f}{\partial u_i \partial u_j} + X_i \frac{\partial Y_j}{\partial x_i} \frac{\partial f}{\partial u_j} \right) - \sum_{i,j} \left(X_i Y_j \frac{\partial^2 f}{\partial u_i \partial u_j} + Y_i \frac{\partial X_j}{\partial u_i} \frac{\partial f}{\partial u_i} \right) \\ &= \sum_{i=1}^n \frac{\partial f}{\partial u_i} \sum_{j=1}^n (X_j \frac{\partial Y_i}{\partial u_j} - Y_j \frac{\partial X_i}{\partial u_j}). \end{aligned}$$

We see that under local coordinates,

$$[X, Y] = (D_X Y_1 - D_Y X_1, \dots, D_X Y_n - D_Y X_n).$$

In particular,

$$[X, Y] = (-z, 0, x) = x \frac{\partial}{\partial z} - z \frac{\partial}{\partial x}.$$

□

- Exercise 2.3.** (1) Let $\mathbb{T}^2$ denote the 2-dimensional torus $S^1 \times S^1$. Construct a vector field $X \in \Gamma(T\mathbb{T}^2)$ that does not have any zeroes.
(2) Construct a vector field $X \in \Gamma(TS^2)$ that has only one zero.

Solution. (1) $X = \frac{\partial}{\partial \theta_1} + \frac{\partial}{\partial \theta_2}$.

(2) S^2 admits two charts on an open cover $S^2 \setminus \{(0, 0, 1)\}, S^2 \setminus \{(0, 0, -1)\}$, given by stereographic projections, with transition functions

$$\begin{aligned}\varphi: (x, y) &\mapsto \left(-\frac{x}{x^2 + y^2}, -\frac{y}{x^2 + y^2}\right), \\ \varphi^{-1}: (u, v) &\mapsto \left(-\frac{u}{u^2 + v^2}, -\frac{v}{u^2 + v^2}\right).\end{aligned}$$

In particular, we have

$$\begin{aligned}\begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} &= \begin{pmatrix} \frac{u^2 - v^2}{(u^2 + v^2)^2} & \frac{2uv}{(u^2 + v^2)^2} \\ \frac{2uv}{(u^2 + v^2)^2} & \frac{v^2 - u^2}{(u^2 + v^2)^2} \end{pmatrix}, \\ \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} &= \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix}^{-1} = \begin{pmatrix} \frac{1}{2}(u^2 - v^2) & uv \\ uv & \frac{1}{2}(v^2 - u^2) \end{pmatrix}.\end{aligned}$$

On $S^2 \setminus \{(0, 0, -1)\}$, $X = \frac{\partial}{\partial x}$ is a non-vanishing vector field. As $\varphi_* X = \frac{1}{2}(u^2 - v^2) \frac{\partial}{\partial u} + uv \frac{\partial}{\partial v}$, X extends to a smooth vector field on S^2 , and $(0, 0, 1)$ is its only zero. □

Exercise 2.4. On the standard unit sphere S^3 in $\mathbb{R}^4$, construct a global frame X, Y, Z for the tangent bundle TS^3 .

Solution. Let x, y, z, w be the coordinate functions of $\mathbb{R}^4$. Consider

$$\begin{aligned}X(x, y, z, w) &= (y, -x, w, -z), \\ Y(x, y, z, w) &= (z, -w, -x, y), \\ Z(x, y, z, w) &= (w, z, -y, -x).\end{aligned}$$

Note that X, Y, Z are tangent to (x, y, z, w) and are orthogonal at each point. Moreover, $|X| = |Y| = |Z| = 1$, so X, Y, Z is indeed a global frame. □

Exercise 2.5. Prove that for any finite-dimensional vector spaces U, V, W , there exists a map $\varphi: U \otimes (V \otimes W) \rightarrow (U \otimes V) \otimes W$ that is an isomorphism and identifies $u \otimes (v \otimes w)$ and $(u \otimes v) \otimes w$.

. For each $u \in U$, $V \times W \rightarrow (U \otimes V) \otimes W, (v, w) \mapsto (u \otimes v) \otimes w$ induces a map $\varphi_u: V \otimes W \rightarrow (U \otimes V) \otimes W$. Then $U \times (V \otimes W) \rightarrow (U \otimes V) \otimes W, (u, v \otimes w) \mapsto \varphi_u(v \otimes w)$ induces a map $\varphi: U \otimes (V \otimes W) \rightarrow (U \otimes V) \otimes W$. Conversely, we can define $\psi: (U \otimes V) \otimes W \rightarrow U \otimes (V \otimes W)$ which maps $(u \otimes v) \otimes w$ to $u \otimes (v \otimes w)$. As ψ is the inverse of φ , φ is an isomorphism. $\square$

Exercise 2.6. Recall that an element $x \in V \otimes W$ is called decomposable if there exist $v \in V$ and $w \in W$ such that $x = v \otimes w$. Suppose V admits a basis $\{e_1, \dots, e_n\}$ and W admits a basis $\{f_1, \dots, f_m\}$. Prove that $x = \sum a_{ij}e_i \otimes f_j$ is decomposable if and only if the matrix (a_{ij}) has rank 1.

Solution. Assume x is decomposable, then

$$x = \left(\sum u_i e_i \right) \otimes \left(\sum v_j f_j \right) = \sum u_i v_j e_i \otimes f_j.$$

It follows that $(a_{ij}) = (u_i v_j)$ has rank 1.

Conversely, if (a_{ij}) has rank 1, then $(a_{ij}) = (u_i v_j)$ and

$$x = \left(\sum u_i e_i \right) \otimes \left(\sum v_j f_j \right).$$

$\square$

Exercise 2.7. For any matrices $A \in \text{GL}(k, \mathbb{R})$ and $B \in \text{GL}(l, \mathbb{R})$, prove

$$\det(A \otimes B) = (\det A)^l (\det B)^k.$$

Solution. Note that $A \otimes B = (A \otimes I_l) \cdot (I_k \otimes B)$, and $\det(A \otimes I_l) = (\det A)^l$, $\det(I_k \otimes B) = (\det B)^k$. $\square$

Exercise 2.8. Recall that on an even-dimensional manifold M , an almost complex structure denoted by J is a smooth family of morphism $J_x: T_x M \rightarrow T_x M$ satisfying $J_x^2 = -\text{id}$. Considering the following $(1, 2)$ -tensor field

$$N_J(X, Y) = [X, Y] + J[JX, Y] + J[X, JY] - [JX, JY]$$

for any $X, Y \in \Gamma(TM)$. A celebrated result from Newlander–Nireberg says that J is integrable if and only if $N_J \equiv 0$. Prove that over a closed surface Σ , any almost complex J is always integrable.

Solution. Obviously, N_J is bilinear and anti-symmetric:

$$N_J(Y, X) = [Y, X] + J[JY, X] + J[Y, JX] - [JY, JX] = -N_J(X, Y).$$

Let f be a smooth function on M .

$$\begin{aligned} N_J(fX, Y) &= [fX, Y] + J[fJX, Y] + J[fX, JY] - [fJX, JY] \\ &= f[X, Y] - (Yf)X + fJ[JX, Y] - (Yf)J(JX) \\ &\quad + fJ[X, JY] - (JYf)JX - f[JX, JY] + (JYf)JX \\ &= fN_J(X, Y). \end{aligned}$$

We see that N_J is indeed a tensor.

On a closed surface, we can choose a local frame X, Y such that $Y = JX$ (in fact, for every non-vanishing local vector field X , JX is linearly independent of X at every point). It suffices to show $N_J(X, Y) = 0$. Indeed,

$$N_J(X, Y) = [X, JX] + J[JX, JX] + J[X, J^2X] - [JX, J^2X] = 0.$$

□

Exercise 2.9. Prove that on any Riemannian manifold (M, g) , there exists a unique connection ∇ satisfying, for any $X, Y, Z \in \Gamma(TM)$,

- (i) (metric compatibility) $D_Z g(X, Y) = g(\nabla_Z X, Y) + g(X, \nabla_Z Y)$.
- (ii) (torsion-free) $\nabla_X Y - \nabla_Y X = [X, Y]$.

Solution. Let $\partial_1, \dots, \partial_n$ be a local frame in local coordinates $x_1, \dots, x_n$. Write $g = g_{ij} dx^i \otimes dx^j$ and $\nabla_{\partial_j} \partial_i = \Gamma_{ij}^k \partial_k$. Torsion free implies $\Gamma_{ij}^k = \Gamma_{ji}^k$. By metric compatibility,

$$\partial_l g_{ij} = g(\nabla_{\partial_l} \partial_i, \partial_j) + g(\partial_i, \nabla_{\partial_l} \partial_j) = \Gamma_{il}^k g_{jk} + \Gamma_{jl}^k g_{ik}.$$

Simiarly, we have

$$\begin{aligned} \partial_i g_{jl} &= \Gamma_{ij}^k g_{kl} + \Gamma_{ik}^l g_{jl}, \\ \partial_j g_{il} &= \Gamma_{jl}^k g_{ik} + \Gamma_{ij}^k g_{kl}. \end{aligned}$$

Thus we obtain

$$\Gamma_{ij}^k = g^{kl} \frac{1}{2} (\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}).$$

Conesequently, such connection is uniquely determined, at least locally. Since we can define this connection locally everywhere, these local connections glue into a global one by the local uniqueness. □

Exercise 2.10. Given a Riemannian manifold (M, g) , prove that for any smooth function $F: M \rightarrow \mathbb{R}$, there exists a unique vector field denoted by ∇F satisfying

$$g(\nabla F, X) = XF$$

for any $X \in \Gamma(TM)$. This vector field is called the gradient of F on M . Also, prove that function F is non-decreasing along ∇F . Finally, work out the explicit formula of ∇F for $F: (\mathbb{R}^2, g) \rightarrow \mathbb{R}$ in polar coordinate (r, θ) , where g is taken as the standard inner product.

Solution. Let $x_1, \dots, x_n$ be local coordinates and $\partial_1, \dots, \partial_n$ be the corresponding local frame of tangent bundle. As g is a metric, ∇F is defined at each point and it suffices to show that ∇F is smooth. Write $X = a^i \partial_i$. Then

$$g_{ij}a^i = \partial_j F$$

and it follows that $a^i = g^{ij} \partial_j F$. $(\nabla F)F = g(\nabla F, \nabla F) \geq 0$, so F is non-decreasing along ∇F .

For $(\mathbb{R}^2, g)$, we have $g = dr \otimes dr + r^2 d\theta \otimes d\theta$, so $g^{rr} = 1, g^{\theta\theta} = \frac{1}{r^2}$. We have

$$\nabla F = \partial_r F \partial_r + \frac{1}{r^2} \partial_\theta F \partial_\theta.$$

□

3 HOMEWORK FOR LECTURE 3

Exercise 3.1. Let V be a vector space with basis $\{e_1, \dots, e_n\}$. Then for a fixed k , prove that

$$\{e_{i_1} \wedge \dots \wedge e_{i_k} \mid 1 \leq i_1 < \dots < i_k \leq n\}$$

form a basis of $\bigwedge^k V$. Therefore, $\dim \bigwedge^k V = \frac{n!}{k!(n-k)!}$.

Solution. Let $L_1, \dots, L_k \in V^*$. Then $(L_1, \dots, L_k)(e_{i_1} \wedge \dots \wedge e_{i_k})$ is well-defined. We see that $\{e_{i_1} \wedge \dots \wedge e_{i_k} \mid 1 \leq i_1 < \dots < i_k \leq n\}$ spans $\bigwedge^k V$. On the other hand, let $f_1, \dots, f_k$ be the dual basis of $e_1, \dots, e_n$. We have

$$(f_{j_1}, \dots, f_{j_k})(e_{i_1} \wedge \dots \wedge e_{i_k}) = \delta_{i_1}^{j_1} \dots \delta_{i_k}^{j_k},$$

which implies $\{e_{i_1} \wedge \dots \wedge e_{i_k} \mid 1 \leq i_1 < \dots < i_k \leq n\}$ is linearly independent. □

Exercise 3.2. Let V be a vector space with basis $\{e_1, \dots, e_n\}$, equipped with an inner product $\langle \cdot, \cdot \rangle$ with signature p negative and $n - p$ positive

$$(-, \dots, -, +, \dots, +).$$

Prove that for the Hodge star operator $*$: $\bigwedge^k V^* \rightarrow \bigwedge^{n-k} V^*$, it satisfies $* \circ * = (-1)^{k(n-k)+p}$ on $\bigwedge^k V^*$.

Solution. Let $e_1, \dots, e_n$ be an oriented orthonormal basis of V . If $\{i_1, \dots, i_k, j_1, \dots, j_{n-k}\} = \{1, \dots, n\}$, then

$$*(e_{i_1} \wedge \dots \wedge e_{i_k}) = \text{sgn}(i_1, \dots, i_k, j_1, \dots, j_{n-k})(-1)^\varepsilon \cdot e_{j_1} \wedge \dots \wedge e_{j_{n-k}},$$

where $\varepsilon = \#\{i_k \mid i_k \leq p\}$. As a consequence, we see that

$$\begin{aligned} & (* \circ *) (e_{i_1} \wedge \dots \wedge e_{i_k}) \\ &= \text{sgn}(i_1, \dots, i_k, j_1, \dots, j_{n-k}) \text{sgn}(i_1, \dots, i_k, j_1, \dots, j_{n-k})(-1)^p \cdot e_{i_1} \wedge \dots \wedge e_{i_k} \\ &= (-1)^{k(n-k)+p} e_{i_1} \wedge \dots \wedge e_{i_k}. \end{aligned}$$

□

Exercise 3.3. Let $\{\varphi_{s,t}\}_{(s,t) \in \mathbb{R}^2}$ be a 2-parametrized group of diffeomorphisms on a manifold M . Consider two vector fields defined via the following equations,

$$\begin{aligned} \frac{\partial \varphi_{s,t}}{\partial t} &= X_s \circ \varphi_{s,t}, \\ \frac{\partial \varphi_{s,t}}{\partial s} &= Y_t \circ \varphi_{s,t}. \end{aligned}$$

Then prove the following equality,

$$\frac{\partial X_s}{\partial s} - \frac{\partial Y_t}{\partial t} = [X_s, Y_t].$$

Solution. It suffices to check this equality locally, so pick local coordinates $u^1, \dots, u^n$. Write $\varphi_{s,t} = \varphi(u, s, t)$ and $X_s = X(u, s, t)$, $Y_t = Y(u, s, t)$, treated as (parametrized) functions taking values in $\mathbb{R}^n$.

$$\begin{aligned} \frac{\partial^2 \varphi}{\partial t \partial s} &= \frac{\partial X(\varphi(u, s, t), s, t)}{\partial s} = \frac{\partial X}{\partial u^i}(\varphi, s, t) \frac{\partial \varphi_i}{\partial s}(\varphi, s, t) + \frac{\partial X}{\partial s}(\varphi, s, t) \\ &= \frac{\partial X}{\partial u^i}(\varphi, s, t) Y_i(\varphi, s, t) + \frac{\partial X}{\partial s}(\varphi, s, t). \end{aligned}$$

Similary, we have

$$\frac{\partial^2 \varphi}{\partial s \partial t} = \frac{\partial Y}{\partial u^i}(\varphi, s, t) X_i(\varphi, s, t) + \frac{\partial Y}{\partial t}(\varphi, s, t).$$

As $\frac{\partial^2 \varphi}{\partial s \partial t} = \frac{\partial^2 \varphi}{\partial t \partial s}$, it follows that

$$\frac{\partial X}{\partial s}(u, s, t) - \frac{\partial Y}{\partial t}(u, s, t) = \frac{\partial Y}{\partial u^i}(u, s, t) X_i(u, s, t) - \frac{\partial X}{\partial u^i}(u, s, t) Y_i(u, s, t) = [X, Y].$$

□

Exercise 3.4. Prove that, for vector fields X, Y on a manifold M , the Lie derivative satisfies $\mathcal{L}_X Y = [X, Y]$.

Solution. It suffices to check this equality locally, so pick local coordinates $u^1, \dots, u^n$. The flow generated by X (at least locally) is defined by

$$\begin{aligned} \frac{\partial \varphi}{\partial t}(u, t) &= X(\varphi(u, t)), \\ \varphi(u, 0) &= u. \end{aligned}$$

Write $Y(u) = Y^i \partial_i|_u$. Then

$$(\varphi_{-t})_* Y(\varphi(u, t)) = Y^i(\varphi(u, t)) \frac{\partial \varphi^j}{\partial u^i}(u, -t) \partial_j|_u.$$

We have

$$\begin{aligned} \frac{d}{dt}(\varphi_{-t})_* Y(\varphi(u, t)) &= \left(\frac{\partial Y^i}{\partial u^k} \frac{\partial \varphi^k}{\partial t} \frac{\partial \varphi^j}{\partial u^i}(u, -t) - Y^i(\varphi(u, t)) \frac{\partial^2 \varphi^j}{\partial u^i \partial t}(u, -t) \right) \partial_j|_u \\ &= \left(\frac{\partial Y^i}{\partial u^k} X^k(\varphi(u, t)) \frac{\partial \varphi^j}{\partial u^i}(u, -t) - Y^i(\varphi(u, t)) \frac{\partial}{\partial u^i}(X^j(\varphi(u, -t))) \right) \partial_j|_u \\ &= \left(\frac{\partial Y^i}{\partial u^k} X^k(\varphi(u, t)) \frac{\partial \varphi^j}{\partial u^i}(u, -t) - Y^i(\varphi(u, t)) \frac{\partial X^j}{\partial u^k} \frac{\partial \varphi^k}{\partial u^i}(u, -t) \right) \partial_j|_u \end{aligned}$$

As $\varphi(u, 0) = u$, $\frac{\partial \varphi^j}{\partial u^i}(u, 0) = \delta_i^j$, and

$$\left. \frac{d}{dt} \right|_{t=0} (\varphi_{-t})_* Y(\varphi(u, t)) = \left(\frac{\partial Y^i}{\partial u^j} X^j - Y^j \frac{\partial X^i}{\partial u^j} \right) \partial_i|_u = (\mathcal{L}_X Y)(u).$$

□

Exercise 3.5. Recall that given a non-degenerate 2-form ω on M , any function $H: M \rightarrow \mathbb{R}$ corresponds to a vector field X_H defined by $-dH = \omega(X_H, -)$. For two functions $H, G: M \rightarrow \mathbb{R}$, define

$$\{H, G\} = \omega(X_H, X_G) = -dH(X_G).$$

Then prove that if ω is closed, then $\{-, -\}$ satisfies the Jacobi identity:

$$\{\{H, G\}, F\} + \{\{G, F\}, H\} + \{\{F, H\}, G\} = 0.$$

Solution. By definition, we have

$$\begin{aligned} \{\{H, G\}, F\} &= -X_F \omega(X_H, X_G) \\ &= -(\mathcal{L}_{X_F} \omega)(X_H, X_G) - \omega(\mathcal{L}_{X_F} X_H, X_G) - \omega(X_H, \mathcal{L}_{X_F} X_G) \\ &= -(\iota_{X_F} d\omega + d\iota_{X_F} \omega)(X_H, X_G) - \omega([X_F, X_H], X_G) - \omega(X_H, [X_F, X_G]) \\ &= -\omega([X_F, X_H], X_G) - \omega(X_H, [X_F, X_G]). \end{aligned}$$

We can derive the similar formula for $\{\{G, F\}, H\}, \{\{F, H\}, G\}$. As a consequence,

$$\{\{H, G\}, F\} + \{\{G, F\}, H\} + \{\{F, H\}, G\} = 2(d\omega)(X_F, X_G, X_H) = 0.$$

□

Exercise 3.6. Consider manifold $\mathbb{R}_{>0}^2$ and $\varphi: \mathbb{R}_{>0}^2 \rightarrow \mathbb{R}_{>0}^2$ defined by

$$F(x, y) = (xy, \frac{y}{x}).$$

Compute the pushforward $\varphi_* X$ for vector field $X = x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y}$. Do the same thing for vector field $Y = y \frac{\partial}{\partial x}$.

Solution. $dF_{(x,y)}(X) = 2xy \frac{\partial}{\partial u} - (\frac{y}{x} + \frac{x}{y}) \frac{\partial}{\partial v}$. Since $F^{-1}(u, v) = (\sqrt{\frac{u}{v}}, \sqrt{uv})$,

$$\varphi_*(X)(u, v) = 2u \frac{\partial}{\partial u} - (v + \frac{1}{v}) \frac{\partial}{\partial v}.$$

$dF_{(x,y)}(Y) = y^2 \frac{\partial}{\partial u} - \frac{y^2}{x^2} \frac{\partial}{\partial v}$, so

$$\varphi_*(Y)(u, v) = uv \frac{\partial}{\partial u} - v^2 \frac{\partial}{\partial v}.$$

□

Exercise 3.7. Consider 1-form $\alpha = xdy$ on $\mathbb{R}^2$ and map $\varphi: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$\varphi(x, y) = (xy, e^{-y}).$$

Compute the pullback $\varphi^*\alpha$. Also, verify in this concrete case that $\varphi^*(d\alpha) = d(\varphi^*\alpha)$.

Solution. $\varphi^*\alpha = xyde^{-y} = -xye^{-y}dy$. We have $d\alpha = dx \wedge dy$ and $d(\varphi^*\alpha) = -ye^{-y}dx \wedge dy$, so

$$\varphi^*(d\alpha) = d(xy) \wedge d(e^{-y}) = (xdy + ydx) \wedge (-e^{-y}dy) = -ye^{-y}dx \wedge dy = d(\varphi^*\alpha).$$

□

Exercise 3.8. Let X be a smooth vector field on M such that $X(p) \neq 0$ at some point $p \in M$.

- (1) Prove that there exists a local chart $(U, \varphi: U \rightarrow V)$ near p , where V is an open subset of $\mathbb{R}^n$ in coordinates $(x_1, \dots, x_n)$, such that within U , we have $\varphi_*(X) = \frac{\partial}{\partial x_1}$.
- (2) Given the following three vector fields on $\mathbb{R}^3$,

$$\begin{aligned} X_1 &= x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x}, \\ X_2 &= y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y}, \\ X_3 &= z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z}. \end{aligned}$$

Near $p = (1, 0, 0)$, is it possible to find a local chart as above such that X_i maps to $\frac{\partial}{\partial x^i}$ for $i = 1, 2, 3$ at the same time? If so, construct such a local chart; if not, please give a justifying reason.

Solution. (1) Under local coordinates $x^1, \dots, x^n$, write $X = a^i \frac{\partial}{\partial x^i}$. Moreover we may assume $p = 0$ and $X(0) = \frac{\partial}{\partial x^1}$. Our strategy is to find a local diffeomorphism $x = F(y)$ such that $F_*(\frac{\partial}{\partial y^1}) = X$. Then F satisfies

$$\frac{\partial F^i}{\partial y^1} \frac{\partial}{\partial x^i} = a^i(F(y)) \frac{\partial}{\partial x^i}.$$

For this purpose, consider the following ODEs

$$\begin{cases} \frac{d}{dt} f^i(t) = a^i(f) \\ f^i(0) = p. \end{cases}$$

By ODE theory, we can find a local smooth solution $f(t; p)$ for p near 0, which is smooth by smooth dependence on initial conditions. Now consider

$$F(y) = f(y^1; 0, y^2, \dots, y^n).$$

We have

$$\frac{\partial F^j}{\partial x^1}(y) = a^j(f(x^1; 0, x^2, \dots, x^n)) = a^j(F(y))$$

and $F(0, y^2, \dots, y^n) = f(0; y^2, \dots, y^n) = (0, y^2, \dots, y^n)$, so F has Jacobian matrix

$$\begin{pmatrix} a^1(F) & a^2(F) & \cdots & a^n(F) \\ & & A & \end{pmatrix},$$

where $A(0) = I_{n-1}$. By inverse function theorem, F is indeed a local diffeomorphism near 0. $\square$

Exercise 3.9. Consider $\mathbb{R}^3$ equipped with the (pseudo)-metric $g = dx \otimes dx + dy \otimes dy - dz \otimes dz$. A Killing vector field on $(\mathbb{R}^3, g)$ is a complete non-trivial vector field such that $\mathcal{L}_X g = 0$. In other words, by the definition of a Lie derivative, the flow generated by X preserves the metric g .

- (1) List as many linearly independent Killing vector fields in $(\mathbb{R}^3, g)$ as possible.
- (2) Verify that if X, Y are two Killing vector fields in $(\mathbb{R}^3, g)$, then $[X, Y]$ is also a Killing vector field in $(\mathbb{R}^3, g)$.

Solution. (1) We have

$$\mathcal{L}_X g(Y, Z) = (\mathcal{L}_X g)(Y, Z) + g(\mathcal{L}_X Y, Z) + g(Y, \mathcal{L}_X Z).$$

On the other hand,

$$\mathcal{L}_X g(Y, Z) = Xg(Y, Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z).$$

Combining the two equalities, we obtain

$$0 = g([X, Y] - \nabla_X Y, Z) + g(Y, [X, Z] - \nabla_X Z) = g(\nabla_Y X, Z) + g(Y, \nabla_Z X).$$

Let $X = X^i \partial_i$ be a Killing vector field on $(\mathbb{R}^3, g)$. Then

$$\begin{aligned} 0 &= g(\nabla_{\partial_i} X, \partial_j) + g(\partial_i, \nabla_{\partial_j} X) \\ &= g(\partial_i X^k \partial_k, \partial_j) + g(\partial_i, \partial_j X^k \partial_k) \\ &= \partial_i X^k g_{jk} + \partial_j X^k g_{ik}. \end{aligned}$$

As $(g_{ij}) = \text{diag}(1, 1, -1)$, $(\partial_i X^k)(g_{jk}) + ((\partial_i X^k)(g_{jk}))^T = 0$, which implies we have a skew-symmetric matrix

$$\begin{pmatrix} \partial_1 X^1 & \partial_1 X^2 & -\partial_1 X^3 \\ \partial_2 X^1 & \partial_2 X^2 & -\partial_2 X^3 \\ \partial_3 X^1 & \partial_3 X^2 & -\partial_3 X^3 \end{pmatrix}.$$

In other words,

$$\begin{cases} \partial_1 X^1 = \partial_2 X^2 = -\partial_3 X^3 = 0 \\ \partial_1 X^2 + \partial_2 X^1 = 0 \\ \partial_1 X^3 = \partial_3 X^1 \\ \partial_3 X^2 = \partial_2 X^3. \end{cases}$$

It follows that $\partial_x, \partial_y, \partial_z, -y\partial_x + x\partial_y, z\partial_y + y\partial_z, z\partial_x + x\partial_z$ form a basis of Killing vector fields in $(\mathbb{R}^3, g)$.

(2) We have $\mathcal{L}_{[X,Y]}g = \mathcal{L}_X(\mathcal{L}_Y g) - \mathcal{L}_Y(\mathcal{L}_X g) = 0$. □

Exercise 3.10. Let α be a 1-form on a 3-dimensional manifold M satisfying $\alpha \wedge d\alpha$ is a nowhere vanishing 3-form on M .

- (1) Prove that there exists a vector field (called the Reeb vector field) denoted by R_α such that $d\alpha(R_\alpha, -) = 0$ and $\alpha(R_\alpha) = 1$.
- (2) Confirm that $\mathcal{L}_{R_\alpha}\alpha = 0$.
- (3) In $\mathbb{R}^3$, give an example of such α and work out the associated R_α .

Solution. (1) Since $\alpha \wedge d\alpha$ is a volume form, we can find a local frame X, Y, Z of TM such that $\alpha \wedge d\alpha(X, Y, Z) = 1$. In particular, we can take X, Y, Z such that $\alpha(X) = 1$ and $\alpha(Y, Z) = 0$. Notice that

$$\alpha \wedge d\alpha(X, Y, Z) = \alpha(X)d\alpha(Y, Z) + \alpha(Y)d\alpha(Z, X) + \alpha(Z)d\alpha(Y, X),$$

hence $d\alpha(Y, Z) = 1$. We can replace X by $X - d\alpha(X, Z)Y + d\alpha(X, Y)Z$, so assume $d\alpha(X, Y, Z) = 0$. Notice that X satisfies the Reeb vector field condition locally. Suppose $R_\alpha = aX + bY + cZ$. Then we have $1 = R_\alpha(X) = a$ and

$$\begin{aligned} 0 &= d\alpha(R_\alpha, Y) = -c, \\ 0 &= d\alpha(R_\alpha, Z) = b. \end{aligned}$$

It follows that R_α is uniquely determined locally, so it is globally defined.

(2) By Cartan's magic formula, $\mathcal{L}_{R_\alpha}\alpha = d\iota_{R_\alpha}\alpha + \iota_{R_\alpha}d\alpha = 0$.

(3) Let $\alpha = dz - ydx$. Then $d\alpha = dx \wedge dy$ and $\alpha \wedge d\alpha = dx \wedge dy \wedge dz$. We see that $\frac{\partial}{\partial z}$ is the Reeb vector field. $\square$

4 HOMEWORK FOR LECTURE 4

Exercise 4.1. Consider unit open disk $B^2 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 < 1\}$ equipped with the following Riemannian metric

$$g = \frac{4}{(1 - (x^2 + y^2))^2} (dx \otimes dx + dy \otimes dy).$$

Meanwhile, consider the open upper half plane $\mathbb{H}^2 = \{(x, y) \in \mathbb{R}^2 \mid y > 0\}$ equipped with the following Riemannian metric

$$g' = \frac{1}{y^2} (dx \otimes dx + dy \otimes dy).$$

Prove that there exists a smooth diffeomorphism $F: B^2 \rightarrow \mathbb{H}^2$ such that it preserves the metrics in the sense that $F^*g' = g$.

Solution. Consider

$$\begin{aligned} F: B^2 \rightarrow \mathbb{H}^2, (x, y) &\mapsto \frac{1}{(1-x)^2 + y^2} (-2y, 1 - x^2 - y^2), \\ G: \mathbb{H}^2 \rightarrow B^2, (u, v) &\mapsto \frac{1}{u^2 + (v+1)^2} (u^2 + v^2 - 1, -2v). \end{aligned}$$

One can check that G is the inverse of F , so F is a diffeomorphism. Meanwhile,

$$\begin{aligned} F^*du &= \frac{4(x-1)y}{(x-1)^2 + y^2} dx + \frac{2y^2 - 2(x-1)^2}{(x-1)^2 + y^2} dy, \\ F^*dv &= \frac{2(x-1)^2 - 2y^2}{(x-1)^2 + y^2} dx + \frac{4(x-1)y}{(x-1)^2 + y^2} dy. \end{aligned}$$

Then

$$\begin{aligned} F^*du \otimes F^*du &= \frac{16(x-1)^2y^2}{((x-1)^2 + y^2)^2} dx \otimes dx \\ &\quad + \frac{8(x-1)y(y^2 - (x-1)^2)}{((x-1)^2 + y^2)^2} (dx \otimes dy + dy \otimes dx) + \frac{4(y^2 - (x-1)^2)^2}{((x-1)^2 + y^2)^2} dy \otimes dy, \\ F^*dv \otimes F^*dv &= \frac{4(y^2 - (x-1)^2)^2}{((x-1)^2 + y^2)^2} dx \otimes dx \\ &\quad - \frac{8(x-1)y(y^2 - (x-1)^2)}{((x-1)^2 + y^2)^2} (dx \otimes dy + dy \otimes dx) + \frac{16(x-1)^2y^2}{((x-1)^2 + y^2)^2} dy \otimes dy. \end{aligned}$$

As a consequence,

$$\begin{aligned}
F^*g' &= \frac{((1-x)^2 + y^2)^2}{(1 - (x^2 + y^2)^2)} \frac{16(x-1)^2y^2 + 4(y^2 - (x-1)^2)^2}{((x-1)^2 + y^2)^2} dx \otimes dx \\
&+ \frac{((1-x)^2 + y^2)^2}{(1 - (x^2 + y^2)^2)} \frac{16(x-1)^2y^2 + 4(y^2 - (x-1)^2)^2}{((x-1)^2 + y^2)^2} dy \otimes dy \\
&= \frac{4}{(1 - (x^2 + y^2))^2} (dx \otimes dx + dy \otimes dy).
\end{aligned}$$

□

Exercise 4.2. Consider map $\Phi: \mathbb{R}^4 \rightarrow \mathbb{R}^2$ defined by

$$\Phi(x, y, s, t) = (x^2 + y, x^2 + y^2 + s^2 + t^2 + y).$$

Show that $(0, 1) \in \mathbb{R}^2$ is a regular value of Φ , and the level set $\Phi^{-1}((0, 1))$ is diffeomorphic to S^2 .

Solution. Φ has Jacobian matrix

$$\begin{pmatrix} 2x & 1 & 0 & 0 \\ 2x & 2y + 1 & 2s & 2t \end{pmatrix}$$

We may assume $s = t = 0$, otherwise (x, y, s, t) must be a regular point. Then $(x, y, 0, 0)$ is critical if and only if $4xy = 0$. As $x^2 + y = 0$, $x^2 + y^2 + y = 1$, we have $y^2 = 1$, which implies $x \neq 0$ and $(0, 1)$ is a regular value.

We have

$$\Phi^{-1}((0, 1)) = \{(x, -x^2, s, t) \mid x^4 + s^2 + t^2 = 1\}.$$

Consider $\pi: \mathbb{R}^4 \rightarrow \mathbb{R}^3$, $(x, y, s, t) \mapsto (x, s, t)$ and $f: \mathbb{R}^3 \rightarrow \mathbb{R}^4$, $(x, s, t) \mapsto (x, -x^2, s, t)$. Let $S = \{(x, s, t) \mid x^4 + y^2 + t^2 = 1\}$, which is a smooth submanifold of $\mathbb{R}^3$. Then $\pi: \Phi^{-1}((0, 1)) \rightarrow S$, $f: S \rightarrow \Phi^{-1}((0, 1))$ show that $\Phi^{-1}((0, 1))$ and S are diffeomorphic. It remains to show that S is diffeomorphic to S^2 . Indeed, we can consider

$$\begin{aligned}
g: S &\rightarrow S^2, (x, s, t) \mapsto (x, \frac{s}{\sqrt{x^2 + 1}}, \frac{t}{\sqrt{x^2 + 1}}), \\
h: S^2 &\rightarrow S, (u, v, w) \mapsto (u, \sqrt{u^2 + 1}v, \sqrt{u^2 + 1}w).
\end{aligned}$$

□

Exercise 4.3. Let N be a non-empty smooth compact manifold. Show that there is no smooth submersion $F: N \rightarrow \mathbb{R}^k$ for any $k > 0$.

Solution. F must be an open map by the canonical submersion theorem. On the other hand, $F(M)$ is compact, so $F(M) = \mathbb{R}^k$, a contradiction. $\square$

Exercise 4.4. Let $N \subset \mathbb{R}^m$ be a smooth submanifold of dimension $n \leq m - 3$. Prove that the complement $\mathbb{R}^m \setminus N$ is simply connected.

Solution. It suffices to show that $\mathbb{R}^m \setminus N$ is path connected. Let $\gamma: I \rightarrow \mathbb{R}^m$ be a smooth path. Then γ is smoothly homotopic to a path γ' intersecting N transversally. For dimension reason $\gamma'(I) \cap N = \emptyset$, so $\gamma'(I) \subset \mathbb{R}^m \setminus N$.

Let $H: I^2 \rightarrow \mathbb{R}^m$ be a homotopy between two smooth paths in $\mathbb{R}^m \setminus N$ (This can be done since $\mathbb{R}^m$ is simply connected). We can further require H to be smooth. Again, H is smoothly homotopic to H' intersecting N transversally. For dimension reason $H'(I^2) \cap N = \emptyset$, so $H'(I) \subset \mathbb{R}^m \setminus N$ and $\mathbb{R}^m \setminus N$ is simply connected. $\square$

Exercise 4.5. Let $F: M \rightarrow M$ be a smooth map. A fixed point $p \in F$ is called non-degenerate if 1 is not an eigenvalue of $dF_p: T_p M \rightarrow T_p M$. The map F is called the Lefschetz map if all its points are non-degenerate.

- (1) Prove that the horizontal rotation $r_\theta: S^2 \rightarrow S^2$ by angle $\theta \neq 2k\pi, k \in \mathbb{Z}$ defined by

$$r_\theta(x, y, z) = (x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta, z)$$

is a Lefschetz map, where S^2 here is viewed as a submanifold in $\mathbb{R}^3$ defined by

$$S^2 = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\}. \quad (4.1)$$

- (2) Let V be a vector space and let $F: V \rightarrow V$ be a linear map. Recall that $\Delta = \{(v, v) \in V \times V \mid v \in V\}$ is the diagonal and $\Gamma_F = \{(v, Fv) \in V \times V \mid v \in V\}$ is the graph of F in $V \times V$. Prove that Γ_F intersects Δ transversally if and only if 1 is not an eigenvalue of F . Then deduce that if M is a compact manifold and $F: M \rightarrow M$ is a Lefschetz map, then there are only finitely many fixed points.
- (3) When M is a compact manifold and $F: M \rightarrow M$ be a Lefschetz map, denote by

$$L(F) = \sum_{F(p)=p} \text{sign}(\det(dF_p - \text{id})).$$

This $L(F)$ is called the Lefschetz number of Lefschetz map F . Compute $L(r_\theta)$ in Question (1) above.

Solution. (1) r_θ has exactly two fixed point $(0, 0, 1), (0, 0, -1)$. Near $(0, 0, 1)$ we have local chart $D^2 \rightarrow S^2, (x, y) \mapsto (x, y, \sqrt{1 - x^2 - y^2})$. Then near $(0, 0, 1)$, r_θ writes $(x, y) \mapsto (x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta)$ and has Jacobian $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ at $(0, 0, 1)$. Similarly, r_θ has Jacobian $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ at $(0, 0, -1)$.

(2) Γ_F intersects Δ transversally if and only if Γ_F, Δ span V , if and only if $\Gamma_F \cap \Delta = 0$. One see that $(v, Fv) \in V \times V$ if and only if $Fv = v$, so the result follows.

If M is compact, $\{\text{fixed points of } F\} = \Gamma_F \cap \Delta$. As F is a Lefschitz map, Γ_F and Δ intersect transversally, so the fixed point set is a zero dimension submanifold of $M \times M$. Moreover, as Γ_F and Δ are both compact, the fixed point set is closed and therefore a finite set in $M \times M$.

(3) We have

$$dF_p - \text{id} = \begin{pmatrix} \cos \theta - 1 & -\sin \theta \\ \sin \theta & \cos \theta - 1 \end{pmatrix}$$

for $p = (0, 0, 1), (0, 0, -1)$. $\det(dF_p - \text{id}) = 2 - 2 \cos \theta > 0$, so $L(F) = 2$. $\square$

Exercise 4.6. Recall that the group of $2n$ symplectic matrices is denoted by

$$\text{Sp}(2n) = \{A \in M(2n, \mathbb{R}) \mid AJ_0A^T = J_0\}$$

where J_0 is denoted by

$$J_0 = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}.$$

Prove that $\text{Sp}(2n)$ is a submanifold of $M(2n, \mathbb{R})$. Moreover, compute its dimension.

Solution. By linear algebra, we have $\text{Sp}(2n) \subset \text{GL}(2n, \mathbb{R})$. Consider $F: \text{GL}(2n, \mathbb{R}) \rightarrow M(2n, \mathbb{R}), A \mapsto AJ_0A^T - J_0$. Then $\text{Sp}(2n) = F^{-1}(0)$ and the differential writes $dF_A: M(2n, \mathbb{R}) \rightarrow M(2n, \mathbb{R}), B \mapsto AJ_0B^T + BJ_0A^T$. Notice that $J_0^T = -J_0$, so for every $A_1, A_2 \in \text{GL}(2n, \mathbb{R})$, $\ker dF_{A_1} \rightarrow \ker dF_{A_2}, B \mapsto BJ_0A_2^T(J_0A_1^T)^{-1}$ is an isomorphism. Therefore, 0 is a regular value of F and $\text{Sp}(2n)$ is a submanifold of $\text{GL}(2n, \mathbb{R}) \subset M(2n, \mathbb{R})$.

$\ker dF_I = \{B \in M(2n, \mathbb{R}) \mid J_0B^T + BJ_0 = 0\}$. Write $B = \begin{pmatrix} B_1 & B_2 \\ B_3 & B_4 \end{pmatrix}$. Then

$$\ker dF_I = \left\{ \begin{pmatrix} B_1 & B_2 \\ B_3 & B_4 \end{pmatrix} \mid B_2 = B_2^T, B_1 + B_4 = 0, B_3 = B_3^T \right\}$$

has dimension $2n^2 + n$. $\square$

Exercise 4.7. Prove by definition that if $N_1 \subset \mathbb{R}^{m_1}$ and $N_2 \subset \mathbb{R}^{m_2}$ are submanifolds of dimension n_1 and n_2 respectively, then $N_1 \times N_2$ is a submanifold of $\mathbb{R}^{m_1+m_2}$ of dimension $n_1 + n_2$.

Solution. Take $p_1 \in N_1$ and $p_2 \in N_2$. Then we can find local charts $\varphi_1: U_1 \rightarrow \mathbb{R}^{m_1}$ and $\varphi_2: U_2 \rightarrow \mathbb{R}^{m_2}$ such that $\varphi_1(N_1 \cap U_1) = \{(x_1, \dots, x_{n_1}, 0^{m_1-n_1})\}$ and $\varphi_2(N_2 \cap U_2) = \{(x_1, \dots, x_{n_2}, 0^{m_2-n_2})\}$. Then

$$\varphi_1 \times \varphi_2: U_1 \times U_2 \rightarrow \mathbb{R}^{n_1+n_2}$$

satisfies $(\varphi_1 \times \varphi_2)((N_1 \times N_2) \cap (U_1 \times U_2)) = \{(x_1, \dots, x_{n_1}, 0^{m_1-n_1}, x'_1, \dots, x'_{n_2}, 0^{m_2-n_2})\}$. $\square$

Exercise 4.8. (1) Prove the Inverse Mapping Theorem: Let $F: N \rightarrow M$ be a smooth map such that $dF_p: T_p N \rightarrow T_{F(p)} M$ is an isomorphism, then F is a diffeomorphism locally near p .

(2) Deduce from (1) that there is no immersion $S^n \rightarrow \mathbb{R}^n$.

Solution. (1) As dF_p is an isomorphism, $\dim M = \dim N$ and the Jacobian matrix of F at every p is invertible. The conclusion follows from Euclidean version of Inverse Mapping Theorem.

(2) As $\dim S^n = \dim \mathbb{R}^n = n$, an immersion $F: S^n \rightarrow \mathbb{R}^n$ must be an local diffeomorphism, so F is an open map. It follows that $F(S^n)$ is both open and compact, so $F(S^n) = \mathbb{R}^n$ is compact, a contradiction. $\square$

Exercise 4.9. Let $F: N \rightarrow M$ be a smooth map. Recall that pullback of F is a functor $F^*: TM \rightarrow TN$. In particular, F^* defines a map for sections from $\Omega^k(M)$ to $\Omega^k(N)$ for any $k \in \mathbb{Z}_{\geq 0}$, defined explicitly as follows,

$$(F^*\alpha)(X_1, \dots, X_k) = \alpha(dF(X_1), \dots, dF(X_k))$$

or even more explicitly when the positions are specified,

$$(F^*\alpha)(p)(X_1, \dots, X_k) = \alpha(F(p))(dF_p(X_1), \dots, dF_p(X_k)).$$

Now consider map $F: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, where $\mathbb{R}^2$ is in coordinate (x, y) and $\mathbb{R}^3$ in coordinate (u, v, w) , by $F(x, y) = (xy, x^2, 3x+y)$. For $\alpha = uvdu + 2w dv - vdw \in \Omega^1(\mathbb{R}^3)$, compute $F^*\alpha$ and express it in terms of dx and dy .

Solution. We have

$$F^* du = xdy + ydx, F^* dv = 2xdx, F^* dw = 3dx + dy,$$

so

$$\begin{aligned} F^* \alpha &= x^3 y(xdy + ydx) + 2(3x + y)(2xdx) - x^2(3dx + dy) \\ &= (x^3 y^2 + 9x^2 + 4xy)dx + (x^4 y - x^2)dy. \end{aligned}$$

□

Exercise 4.10. For map $F: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by

$$F(x, y) = (e^y \cos x, e^y \sin x, e^{-y}).$$

Denote by $S(0, r) \subset \mathbb{R}^3$ the standard 2-sphere centered at 0 with radius r . Recall that a map $F: N \rightarrow M$ is transversal to a submanifold $S \subset M$ means for any $x \in F^{-1}(S)$, linear spaces $T_{F(x)}S$ and $dF_x(T_x N)$ span $T_{F(x)}M$.

- (1) For which positive number r is F transversal to $S(0, r)$ in $\mathbb{R}^3$?
- (2) For which positive number r is $F^{-1}(S(0, r))$ an embedded submanifold of $\mathbb{R}^2$?

Solution. (1) We have

$$F^{-1}(S(0, r)) = \{(x, y) \in \mathbb{R}^2 \mid e^{2y} + e^{-2y} = r^2\}.$$

F has Jacobian matrix

$$\begin{pmatrix} -e^y \sin x & e^y \cos x & 0 \\ e^y \cos x & e^y \sin x & -e^{-y} \end{pmatrix}.$$

$dF(\partial_x) = -e^y \sin x \partial_u + e^y \cos x \partial_v$, $dF(\partial_y) = e^y \cos x \partial_u + e^y \sin x \partial_v - e^{-y} \partial_w$. On the other hand, $T_{F(x)}S$ has normal space spanned by $n = e^y \cos x \partial_u + e^y \sin x \partial_v + e^{-y} \partial_w$. Then F is transversal to $S(0, r)$ if and only if

$$(\langle -e^y \sin x \partial_u + e^y \cos x \partial_v, n \rangle, \langle e^y \cos x \partial_u + e^y \sin x \partial_v - e^{-y} \partial_w, n \rangle) \neq 0,$$

if and only if $e^{2y} - e^{-2y} \neq 0$. In other words, F is transversal to $S(0, r)$ if and only if $r \neq \sqrt{2}$.

(2) If $r \neq \sqrt{2}$, then $F^{-1}(S(0, r))$ is an embedded submanifold. If $r = \sqrt{2}$, $F^{-1}(S(0, r)) = \{(x, y) \mid y = 0\}$, which is also an embedded submanifold. We see that $F^{-1}(S(0, r))$ is always an embedded submanifold. □

5 HOMEWORK FOR LECTURE 5

Exercise 5.1. Let M be a smooth manifold and $F: M \rightarrow \mathbb{R}^k$ be a continuous map. Prove that for any positive continuous function $\varepsilon: M \rightarrow \mathbb{R}$, there exists a smooth map $G: M \rightarrow \mathbb{R}^k$ such that $\|G(x) - F(x)\| \leq \varepsilon(x)$ for any $x \in M$.

Solution. Since the requirement is pointwise, we may assume $M = \mathbb{R}^n$ (by taking a P.O.U. with respect to a chart cover). Let $\{U_i\}$ be a precompact open cover and take a P.O.U. $\{\rho_i\}$ with respect to $\{U_i\}$. Let $\varphi_i = \inf_{U_i} \varepsilon > 0$. For every $f \in C^\infty(\mathbb{R}^n)$, we can find $g_i \in C^\infty(U_i)$ such that $\sup_{U_i} \|f - g_i\| \leq \frac{\varepsilon_i}{\sqrt{n}}$. Let $g = \sum \rho_i g_i$. Then

$$\|f(x) - g(x)\| = \left\| \sum \rho_i(x) (f(x) - g_i(x)) \right\| \leq \sum \rho_i \frac{\varepsilon_i}{\sqrt{n}} \leq \frac{\varepsilon(x)}{\sqrt{n}}.$$

Now suppose $F = (F_1, \dots, F_k)$. Then pick $G_m \in C^\infty(\mathbb{R}^n)$ as above for every F_m . We have

$$\|G - F\| = \left(\sum_{m=1}^k (G_m(x) - F_m(x))^2 \right)^{\frac{1}{2}} \leq \varepsilon(x).$$

□

Exercise 5.2. Consider $\theta \in \Omega^2(\mathbb{R}^3)$ defined by

$$\theta = x^2 dy \wedge dz + y dz \wedge dx + z dx \wedge dy.$$

Denote by $S^2 = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\}$. Compute the integration $\int_{S^2} \iota^* \theta$ where $\iota: S^2 \rightarrow \mathbb{R}^3$ is the inclusion.

Solution. Let B^3 be the unit ball in $\mathbb{R}^3$. Then $\partial B^3 = S^2$, so by the Stokes' formula, we have

$$\theta \in \Omega^2(\mathbb{R}^3) = \int_{B^3} d\theta = \int_{B^3} (2x + 2) dx \wedge dy \wedge dz = 2 \text{vol}(B^3) = \frac{8\pi}{3}.$$

□

Exercise 5.3. (1) Given a manifold M and two 1-forms $\alpha, \beta \in \Omega^1(M)$, prove the following identity

$$\begin{aligned} & \alpha \wedge (d\alpha)^n - \beta \wedge (d\beta)^n \\ &= (\alpha - \beta) \wedge \sum_{j=0}^n ((d\alpha)^j \wedge (d\beta)^{n-j}) + d \left(\alpha \wedge \beta \wedge \sum_{j=0}^{n-1} (d\alpha)^j \wedge (d\beta)^{n-1-j} \right) \end{aligned}$$

for any $n \in \mathbb{Z}_{\geq 0}$.

- (2) Deduce the following proposition from (1) in this exercise: Given a closed (i.e., compact without boundary) orientable manifold M of dimension $2n + 1$ and smooth vector field $X \in \Gamma(TM)$, if two 1-forms $\alpha, \beta \in \Omega^1(M)$ satisfy $(\phi_X^t)^*\alpha = \alpha$ and $(\phi_X^t)^*\beta = \beta$ for any $t \in \mathbb{R}$, moreover, $\alpha(X) = \beta(X) = 1$, then

$$\int_M \alpha \wedge (d\alpha)^n = \int_M \beta \wedge (d\beta)^n.$$

Note that the invariant condition above can also be expressed as $\mathcal{L}_X \alpha = \mathcal{L}_X \beta = 0$.

Solution. (1) We have

$$\begin{aligned} & d \left(\alpha \wedge \beta \wedge \sum_{j=0}^{n-1} (d\alpha)^j \wedge (d\beta)^{n-1-j} \right) \\ &= (d\alpha) \wedge \beta \wedge \sum_{j=0}^{n-1} (d\alpha)^j \wedge (d\beta)^{n-1-j} - \alpha \wedge (d\beta) \wedge \sum_{j=0}^{n-1} (d\alpha)^j \wedge (d\beta)^{n-1-j} \\ &= -\beta \wedge \sum_{j=0}^{n-1} (d\alpha)^{j+1} \wedge (d\beta)^{n-1-j} + \alpha \wedge \sum_{j=0}^{n-1} (d\alpha)^j \wedge (d\beta)^{n-j}. \end{aligned}$$

(2) By Stokes' formula and proposition from (1),

$$\int_M \alpha \wedge (d\alpha)^n - \int_M \beta \wedge (d\beta)^n = \int_M (\alpha - \beta) \wedge \sum_{j=0}^n ((d\alpha)^j \wedge (d\beta)^{n-j}).$$

By Cartan's magic formula, we have $0 = \mathcal{L}_X \alpha = \iota_X d\alpha + d(\iota_X \alpha) = \iota_X d\alpha$. Similarly, $\iota_X d\beta = 0$, hence

$$\begin{aligned} & \iota_X \left((\alpha - \beta) \wedge \sum_{j=0}^n ((d\alpha)^j \wedge (d\beta)^{n-j}) \right) \\ &= (\iota_X (\alpha - \beta) \wedge \sum_{j=0}^n ((d\alpha)^j \wedge (d\beta)^{n-j})) \\ &\quad - (\alpha - \beta) \wedge \sum_{j=0}^n \left(\sum_{k=0}^{j-1} (d\alpha)^k \wedge (\iota_X d\alpha) \wedge (d\alpha)^{j-k-1} \wedge (d\beta)^{n-j} \right) \\ &\quad - (\alpha - \beta) \wedge \sum_{j=0}^n \left(\sum_{k=0}^{n-j-1} (d\alpha)^j \wedge (d\beta)^k \wedge (\iota_X d\beta) \wedge (d\beta)^{n-j-k-1} \right) \\ &= 0. \end{aligned}$$

Take an arbitrary top form $2n + 1$ -form ω . Notice that $\alpha \wedge \omega = 0$, so $0 = \iota_X(\alpha \wedge \omega) = \alpha(X)\omega - \alpha \wedge \iota_X\omega$. In particular, $\omega = \alpha \wedge \iota_X\omega$. It follows that $\omega = 0$ if and only if $\iota_X\omega = 0$, so the conclusion follows. $\square$

Exercise 5.4. Let M be a closed manifold of dimension $2n$.

- (1) Let $\omega \in \Omega^2(M)$ be a 2-form, then ω is non-degenerate (in the sense that at any point $x \in M$, if $v \in T_xM$ is not zero, then there exists some $w \in T_xM$ such that $\omega_x(v, w) \neq 0$) if and only if ω^n is a volume form of M .
- (2) From Exercise 3.5, we have seen the Poisson bracket of two functions $H, G: M \rightarrow \mathbb{R}$ defined by

$$\{H, G\} : M \rightarrow \mathbb{R}$$

defined by

$$\{H, G\} = \omega(X_H, X_G),$$

where $-dH = \omega(X_H, -)$. Suppose further that ω is closed, then prove that

$$\int_M \{F, G\} \omega^n = 0.$$

Solution. As $dG \wedge \omega^n = 0$,

$$0 = \iota_{X_F}(dG \wedge \omega^n) = dG(X_F)\omega^n - dG \wedge \iota_{X_F}\omega^n = \{F, G\} \omega^n + dG \wedge dF \wedge \omega^{n-1}.$$

We obtain $\{F, G\} \omega^n = dF \wedge dG \wedge \omega^{n-1} = d(FdG \wedge \omega^{n-1})$, so the conclusion follows from Stokes' formula. $\square$

Exercise 5.5. Let M, N be orientable manifolds and $\dim M = m, \dim N = n$. Let $\pi_M: M \times N \rightarrow M$ and $\pi_N: M \times N \rightarrow N$ be the projections. Then for forms $\alpha \in \Omega^m(M)$ and $\beta \in \Omega^n(N)$, consider their product defined by

$$\alpha \times \beta = \pi_M^* \alpha \wedge \pi_N^* \beta \in \Omega^{m+n}(M \times N)$$

prove from definition that

$$\int_{M \times N} \alpha \times \beta = \left(\int_M \alpha \right) \cdot \left(\int_N \beta \right).$$

Solution. Take a chart cover $\{U_i\}, \{V_j\}$ of M, N respectively and a P.O.U. $\{\rho_i\}, \{\eta_j\}$ with respect to $\{U_i\}, \{V_j\}$. Then $\{\rho_i \eta_j\}$ is a P.O.U. with respect to $\{U_i \times V_j\}$.

$$\begin{aligned} \int_{M \times N} \alpha \times \beta &= \sum_{i,j} \int_{U_i \times V_j} (\rho_i \eta_j) \pi_M^* \alpha \wedge \pi_N^* \beta = \sum_{i,j} \left(\int_{U_i} \rho_i \alpha \right) \cdot \left(\int_{V_j} \eta_j \beta \right) \\ &= \left(\sum_i \int_{U_i} \rho_i \alpha \right) \cdot \left(\sum_j \int_{V_j} \eta_j \beta \right) = \left(\int_M \alpha \right) \cdot \left(\int_N \beta \right). \end{aligned}$$

□

6 HOMEWORK FOR LECTURE 6

Exercise 6.1. For the following matrix groups $\mathrm{SL}(n, \mathbb{R}), \mathrm{O}(n), \mathrm{SL}(n, \mathbb{C}), \mathrm{U}(n)$ and $\mathrm{Sp}(2n)$, confirm their induced Lie algebras as follows.

- (i) $\mathfrak{sl}(n, \mathbb{R}) = \{A \in \mathfrak{gl}(n, \mathbb{R}) \mid \mathrm{tr} A = 0\}$.
- (ii) $\mathfrak{sl}(n, \mathbb{C}) = \{A \in \mathfrak{gl}(n, \mathbb{C}) \mid \mathrm{tr} A = 0\}$.
- (iii) $\mathfrak{o}(n) = \{A \in \mathfrak{gl}(n, \mathbb{R}) \mid A^T + A = 0\}$.
- (iv) $\mathfrak{u}(n) = \{A \in \mathfrak{gl}(n, \mathbb{C}) \mid A^H + A = 0\}$.
- (v) $\mathfrak{sp}(2n) = \{A \in \mathfrak{gl}(n, \mathbb{R}) \mid A^T J + J A = 0\}$.

Solution. Let $A \in M(n, K)$, where $K = \mathbb{R}, \mathbb{C}$ and suppose A is similar to an upper triangular matrix P , with eigenvalues $\lambda_1, \dots, \lambda_n$. Then

$$\det(I_n + tA) = \det(I_n + tP) = (1 + \lambda_1 t) \cdots (1 + \lambda_n t).$$

It follows that $(d \det)_{I_n}(A) = \mathrm{tr} A$, $A \in \mathfrak{gl}(n, K)$, so we have confirmed (i) and (ii).

Recall that $\mathrm{O}(n) = F^{-1}(I_n)$, where $F: \mathrm{GL}(n, \mathbb{R}) \rightarrow M(n, \mathbb{R}), B \mapsto B^T B$. Since $(I_n + tA)^T(I_n + tA) = I_n + t(A^T + A) + t^2 A^T A$, we have $(dF)_{I_n}(A) = A^T + A$. Similarly, $\mathrm{U}(n) = F^{-1}(I_n)$, where $G: \mathrm{GL}(n, \mathbb{C}) \rightarrow M(n, \mathbb{C}), B \mapsto B^H B$, and we see that $(I_n + tA)^H(I_n + tA) = I_n + t(A^H + A) + t^2 A^H A$, so $(dG)_{I_n}(A) = A^H + A$. Arguments above verify (iii) and (iv).

By Exercise 4.6, we can see that $\mathfrak{sp}(2n)$, the kernel of differential map at identity element, is the space $\{A \in \mathfrak{gl}(n, \mathbb{R}) \mid A^T J + J A = 0\}$. □

Exercise 6.2. Given a Lie group G , prove the following equality:

$$\exp(-tX) \exp(-tY) \exp(tX) \exp(tY) = \exp(t^2[X, Y] + O(t^3))$$

for any $X, Y \in \mathfrak{g}_G$.

Solution. Let $f \in C^\infty(G)$, $Z \in \mathfrak{g}_G$, $g \in G$ and consider $f(\exp(tZ))$. We have

$$\frac{d}{dt}f(g \exp(tZ)) = (df)\left(\frac{d}{dt}(g \exp(tZ))\right) = (\tilde{Z}f)(g \exp(tZ)),$$

where $\tilde{Z}$ is the corresponding left invariant vector field. Consequently, we have $\frac{d^2}{dt^2}f(g \exp(tZ)) = (\tilde{Z}^2f)(g \exp(tZ))$. It follows that

$$\begin{aligned} f(\exp(tX) \exp(sY)) &= \sum_{i=0}^2 \frac{s^i}{i!} (\tilde{Y}^i f)(\exp(tX)) + O(s^3) \\ &= \sum_{i=0}^2 \sum_{j=0}^2 \frac{s^i t^j}{i! j!} (\tilde{X}^j \tilde{Y}^i f)(e) + O(s^3) + O(t^3), \end{aligned}$$

where $O(s^3)$ is uniform with respect to t by the standard mathematical analysis. Therefore,

$$f(\exp(tX) \exp(tY)) = f(e) + t((\tilde{X} + \tilde{Y})f)(e) + \frac{1}{2}t^2((\tilde{X}^2 + 2\tilde{X}\tilde{Y} + \tilde{Y}^2)f)(e) + O(t^3).$$

Denote $W(t) = \log(\exp(tX) \exp(tY))$ and $W(t) = tW_1 + t^2W_2 + O(t^3)$. Then under local coordinates by $\log$ we have

$$\begin{aligned} f(\exp W(t)) &= f(e) + (W(t)f)(e) + \frac{1}{2}(W(t)^2f)(e) + O(t^3) \\ &= f(e) + t(\tilde{W}_1f)(e) + \frac{1}{2}t^2((\tilde{W}_1^2 + 2\tilde{W}_2)f)(e) + O(t^3). \end{aligned}$$

Therefore,

$$\begin{aligned} \tilde{W}_1 &= \tilde{X} + \tilde{Y}, \\ \tilde{W}_1^2 + 2\tilde{W}_2 &= \tilde{X}^2 + 2\tilde{X}\tilde{Y} + \tilde{Y}^2, \end{aligned}$$

hence $2\tilde{W}_2 = \tilde{X}^2 + 2\tilde{X}\tilde{Y} + \tilde{Y}^2 - (\tilde{X} + \tilde{Y})^2 = \tilde{X}\tilde{Y} - \tilde{Y}\tilde{X} = [\tilde{X}, \tilde{Y}]$. In particular, $W_2 = \frac{1}{2}[X, Y]$ and we obtain the following asymptotic expression:

$$\exp(tX) \exp(tY) = \exp\left(t(X + Y) + \frac{1}{2}t^2[X, Y] + O(t^3)\right).$$

The conclusion naturally follows. Indeed,

$$\begin{aligned}
& \exp(-tX) \exp(-tY) \exp(tX) \exp(tY) \\
&= \exp\left(-t(X+Y) + \frac{1}{2}t^2[X, Y] + O(t^3)\right) \exp\left(t(X+Y) + \frac{1}{2}t^2[X, Y] + O(t^3)\right) \\
&= \exp\left(t(-(X+Y) + \frac{1}{2}t[X, Y] + O(t^2))\right) \exp\left(t((X+Y) + \frac{1}{2}t[X, Y] + O(t^2))\right) \\
&= \exp\left(t(t[X, Y] + O(t^2)) + O(t^3)\right) \\
&= \exp\left(t^2[X, Y] + O(t^3)\right).
\end{aligned}$$

□

Exercise 6.3. Prove that the matrix exponential map on elements in $M(n, \mathbb{R})$ satisfies

$$\det(e^A) = e^{\text{tr } A}.$$

Here, $e^A = I_n + A + \frac{A^2}{2} + \dots$. Use this conclusion to confirm that the following matrix

$$\begin{pmatrix} -2 & 0 \\ 0 & -1 \end{pmatrix}$$

can not be written as e^A for any $A \in M(n, \mathbb{R})$.

Solution. Let $A = P^{-1}BP$ where $P \in \text{GL}(n, \mathbb{R})$ and B is the Jordan form. Then $\text{tr } A = \text{tr } B$ and $e^A = P^{-1}e^B P$. It follows that

$$\det(e^A) = \det(e^B) = e^{\text{tr } B} = e^{\text{tr } A}.$$

Moreover, we see that eigenvalues of such matrix are all positive. □

Exercise 6.4. Given a Riemannian metric g , recall that the associated curvature tensor (as a (0,4)-tensor) is defined by

$$R(X, Y, Z, W) = g(R(X, Y)Z, W),$$

where $R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z$. Prove the following equalities.

- (i) $R(X, Y, Z, W) + R(Y, Z, X, W) + R(Z, X, Y, W) = 0$.
- (ii) $R(X, Y, Z, W) = -R(Y, X, Z, W) = -R(X, Y, W, Z)$.
- (iii) $R(X, Y, Z, W) = R(Z, W, X, Y)$.

Solution. (i) Let S be the operator such that:

$$ST(X, Y, Z) = T(X, Y, Z) + T(Y, Z, X) + T(Z, X, Y).$$

Then $R(X, Y, Z, W) + R(Y, Z, X, W) + R(Z, X, Y, W) = g(SR(X, Y)Z, W)$.

$$\begin{aligned} SR(X, Y)Z &= S(\nabla_X \nabla_Y Z) - S(\nabla_Y \nabla_X Z) - S(\nabla_{[X, Y]} Z) \\ &= S(\nabla_Z \nabla_X Y) - S(\nabla_Z \nabla_Y X) - S(\nabla_{[X, Y]} Z) \\ &= S(\nabla_Z [X, Y]) - S(\nabla_{[X, Y]} Z) \\ &= S([Z, [X, Y]]) = 0. \end{aligned}$$

Since $R(Y, X)Z = -R(X, Y)Z$, we have $R(X, Y, Z, W) = -R(Y, X, Z, W)$. $\square$

Exercise 6.5. Consider the following 2-dimensional Lie group

$$G = \left\{ \begin{pmatrix} x & y \\ 0 & 1 \end{pmatrix} \mid x > 0, y \in \mathbb{R} \right\}.$$

Complete the following questions.

(i) Verify that its Lie algebra is

$$\mathfrak{g}_G = \left\{ \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} \mid a, b \in \mathbb{R} \right\}.$$

(ii) Take the following basis of $\mathfrak{g}_G$ in (i),

$$X_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, X_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Construct a left invariant metric g on G such that $\{X_1, X_2\}$ form an orthonormal basis.

(iii) Verify that the corresponding Levi-Civita connection ∇ of g in (ii) satisfies the following relations,

$$\begin{aligned} \nabla_{X_1} X_1 &= \nabla_{X_1} X_2 = 0, \\ \nabla_{X_2} X_1 &= -X_2, \nabla_{X_2} X_2 = X_1. \end{aligned}$$

Here X_1, X_2 are extended to left invariant vector fields on G .

(iv) Compute sectional curvatures of (G, g, ∇) for g and ∇ in (ii) and (iii).

Solution. (i) $\mathfrak{g}_G = T_e G$, so the result follows.

(ii) For $g = \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}$, we have

$$L_g: \begin{pmatrix} x & y \\ 0 & 1 \end{pmatrix} \mapsto \begin{pmatrix} ax & ay + b \\ 0 & 1 \end{pmatrix}.$$

We see that $x^{-2}(dx \otimes dx + dy \otimes dy)$ is a left invariant metric on G , with $\{X_1, X_2\}$ being an orthonormal basis.

(iii) We have $X_1 = x\partial_1, X_2 = x\partial_2$. Write $\nabla_{\partial_i}\partial_j = \Gamma_{ij}^k\partial_k$. We have

$$\Gamma_{ij}^k = g^{kl} \frac{1}{2} (\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}).$$

$(g_{ij}) = \text{diag}(x^{-2}, x^{-2})$ and $(g^{ij}) = \text{diag}(x^2, x^2)$, so

$$\begin{aligned} \Gamma_{11}^1 &= \frac{1}{2} g^{11} (\partial_1 g_{11} + \partial_1 g_{11} - \partial_1 g_{11}) = -\frac{1}{x}, \\ \Gamma_{11}^2 &= \frac{1}{2} g^{22} (\partial_1 g_{12} + \partial_1 g_{12} - \partial_2 g_{11}) = 0, \\ \Gamma_{12}^1 &= \frac{1}{2} g^{11} (\partial_1 g_{12} + \partial_2 g_{11} - \partial_1 g_{12}) = 0, \\ \Gamma_{12}^2 &= \frac{1}{2} g^{22} (\partial_1 g_{22} + \partial_2 g_{12} - \partial_2 g_{12}) = -\frac{1}{x}, \\ \Gamma_{22}^1 &= \frac{1}{2} g^{11} (\partial_2 g_{12} + \partial_2 g_{12} - \partial_1 g_{22}) = \frac{1}{x}, \\ \Gamma_{22}^2 &= \frac{1}{2} g^{22} (\partial_2 g_{22} + \partial_2 g_{22} - \partial_2 g_{22}) = 0. \end{aligned}$$

It follows that

$$\begin{aligned} \nabla_{X_1} X_1 &= x \nabla_{\partial_1} (x \partial_1) = x^2 \nabla_{\partial_1} \partial_1 + x (\partial_1 x) \partial_1 = x \partial_1 - x \partial_1 = 0, \\ \nabla_{X_1} X_2 &= x \nabla_{\partial_1} (x \partial_2) = x^2 \nabla_{\partial_1} \partial_2 + x (\partial_1 x) \partial_2 = x \partial_2 - x \partial_2 = 0, \\ \nabla_{X_2} X_1 &= x \nabla_{\partial_2} (x \partial_1) = x^2 \nabla_{\partial_2} \partial_1 + x (\partial_2 x) \partial_1 = -x \partial_2 = -X_2, \\ \nabla_{X_2} X_2 &= x \nabla_{\partial_2} (x \partial_2) = x^2 \nabla_{\partial_2} \partial_2 + x (\partial_2 x) \partial_2 = X_1. \end{aligned}$$

(iv) The sectional curvature, which is the Gaussian curvature of G , is

$$\begin{aligned} K(p) &= \langle R(X_1, X_2)X_2, X_1 \rangle = \langle \nabla_{X_1} \nabla_{X_2} X_2 - \nabla_{X_2} \nabla_{X_1} X_2 - \nabla_{[X_1, X_2]} X_2, X_1 \rangle \\ &= \langle \nabla_{X_1} X_1 - \nabla_{X_2} X_2, X_1 \rangle = -\langle X_1, X_1 \rangle = -1. \end{aligned}$$

□

7 HOMEWORK FOR LECTURE 7

Exercise 7.1. Prove that any short exact sequence of cochain complexes (of k -modules) $0 \rightarrow (C^\star, d_C) \rightarrow (D^\star, d_D) \rightarrow (E^\star, d_E) \rightarrow 0$ induces a long exact sequence on cohomology groups,

$$\cdots \rightarrow H^*(C^\star; k) \rightarrow H^*(D^\star; k) \rightarrow H^*(E^\star; k) \rightarrow H^{*+1}(C^\star; k) \rightarrow \cdots.$$

Solution. Denote $0 \rightarrow (C^\star, d_C) \xrightarrow{f} (D^\star, d_D) \xrightarrow{g} (E^\star, d_E) \rightarrow 0$.

We first construct the connecting map r . Given a class $[\alpha] \in H^n(E^\star)$. Since $g: D^n \rightarrow E^n$ is surjective, $\alpha = g(\beta)$ for some $\beta \in D^n$. Consider $d\beta$. Since $g(d\beta) = d(g(\beta)) = d\alpha = 0$, there exists a unique element $\gamma \in C^{n+1}$ such that $f(\gamma) = d\beta$. Since $f(d\gamma) = d(f(\gamma)) = d^2\beta = 0$, we have $d\gamma = 0$, which implies $\gamma \in H^{n+1}(D^\star)$. This can be illustrated by the following diagram:

$$\begin{array}{ccccc} & & \beta & \longrightarrow & \alpha \\ & & \downarrow & & \downarrow \\ \gamma & \longrightarrow & d\beta & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \\ d\gamma & \longrightarrow & 0 & & \end{array}$$

Chain maps f, g induces maps $p: H^*(C^\star) \rightarrow H^*(D^\star), q: H^*(D^\star) \rightarrow H^*(E^\star)$ respectively. It remains to show that p, q, r give the long exact sequence.

Since $q \circ p = 0$, naturally $q \circ p = 0$. Conversely, suppose $q[\beta] = [g\beta] = 0$. Then $g\beta = d\gamma$ for some $\gamma \in E^{n-1}$. As g is surjective, $\gamma = g\beta'$ for some $\beta' \in D^{n-1}$ and $\beta - d\beta' \in \ker g = \text{im } f$. We see that $[\beta] = [\beta - d\beta'] \in \text{im } p$.

Take $[\beta] \in H^n(D^\star)$. We have $d\beta = 0$, so by the definition of r , the unique $\gamma \in C^{n+1}$ is zero and $(r \circ q)[\beta] = 0$. On the other hand, for every $[\alpha] \in \ker r$, let β, γ be as in the definition of r . Then $d\beta = 0$ and $[\alpha] = [g\beta] \in \text{im } q$.

Take $[\alpha] \in H^n(E^\star)$ and let β, γ be as in the definition of r . Then $(p \circ r)[\alpha] = p[\gamma] = [f\gamma] = [fd\beta] = 0$. Now suppose $[\eta] \in \ker p$. Then $f\eta = d\beta$ for some $\beta \in D^{n-1}$. It follows that $[\eta] = r[g\beta]$. $\square$

Exercise 7.2. Prove Künneth formula of de Rham cohomology groups. Explicitly, for manifolds M, N with finite-dimensional cohomology,

$$H_{\text{dR}}^k(M \times N) \simeq \bigoplus_{p+q=k} H_{\text{dR}}^p(M) \otimes_{\mathbb{R}} H_{\text{dR}}^q(N).$$

Solution. By universal coefficient theorem, $H_i(M; \mathbb{R})$ and $H_i(N; \mathbb{R})$ are both finitely generated. Hence the general Künneth formula in algebraic topology gives us a short exact sequence

$$0 \rightarrow \bigoplus_{p+q=k} H_p(M; \mathbb{R}) \otimes_{\mathbb{R}} H_q(N; \mathbb{R}) \rightarrow H_k(M \times N; \mathbb{R}) \rightarrow \bigoplus_{p+q=n-1} \text{Tor}(H_p(M; \mathbb{R}), H_q(N; \mathbb{R})) \rightarrow 0.$$

As $\mathbb{R}$ is a field, the last part in the short exact sequence vanishes. The conclusion follows from $H_i(M; \mathbb{R}) \simeq H^i(M; \mathbb{R})$ and de Rham theorem. $\square$

Exercise 7.3. Compute de Rham cohomology groups of $\mathbb{RP}^n$ using Mayer–Vietoris sequence.

Solution. Let S^n be the unit sphere in $\mathbb{R}^{n+1}$ and let

$$X = \{(x^1, \dots, x^{n+1}) \in S^n \mid x^{n+1} \geq 0\}$$

be the northern hemisphere. Denote by S^{n-1} the equator in S^n , which serves as the boundary of X . Then $\mathbb{RP}^n = X / \sim$, where $x \in S^{n-1}$ is identified with $-x$. Take an open cover $\{U, V\}$ of $\mathbb{RP}^n$ as follows,

$$U = \{[(x^1, \dots, x^{n+1})] \in X \mid x^{n+1} > 0\} / \sim, \\ V = \{[(x^1, \dots, x^{n+1})] \in X \mid x^{n+1} < 1\} / \sim.$$

$U = X \setminus S^{n-1}$ is diffeomorphic to $\mathbb{R}^n$ and $U \cap V$ is diffeomorphic to $S^{n-1} \times (0, 1)$. It remains to analyse V . Define

$$p: V \rightarrow S^{n-1}, [(x^1, \dots, x^{n+1})] \mapsto \left[\frac{x^1}{\sqrt{1 - (x^{n+1})^2}}, \dots, \frac{x^n}{\sqrt{1 - (x^{n+1})^2}}, 0 \right].$$

p turns out to be a deformation retraction $\text{rel } S^{n-1} / \sim = \mathbb{RP}^{n-1}$. Indeed, $p(x) = x$ for $x \in \mathbb{RP}^{n-1}$ and

$$H: V \times I, [(x^1, \dots, x^{n+1})] \mapsto \left[\sqrt{1 - t^2} \frac{x^1}{\sqrt{1 - (x^{n+1})^2}}, \dots, \sqrt{1 - t^2} \frac{x^n}{\sqrt{1 - (x^{n+1})^2}}, t \right]$$

is a homotopy between p and id_V . As a consequence, V is homotopy equivalent to $\mathbb{RP}^{n-1}$. Applying Mayer–Vietoris sequence, we have the following exact sequence

$$H_{\text{dR}}^{k-1}(S^{n-1}) \rightarrow H_{\text{dR}}^k(\mathbb{RP}^n) \rightarrow H_{\text{dR}}^k(\mathbb{R}^n) \oplus H_{\text{dR}}^k(\mathbb{RP}^{n-1}) \rightarrow H_{\text{dR}}^k(S^{n-1}).$$

$\mathbb{RP}^n$ is orientable if and only if n is odd, which implies $H_{\text{dR}}^n(\mathbb{RP}^n) = \mathbb{R}$ for n odd and 0 for n even. Recall that $\mathbb{RP}^1 = S^1$, so we have

$$\begin{aligned} 0 \rightarrow H_{\text{dR}}^0(\mathbb{RP}^2) = \mathbb{R} \rightarrow H_{\text{dR}}^0(\mathbb{R}) \oplus H_{\text{dR}}^0(\mathbb{RP}^1) = \mathbb{R}^2 \rightarrow H_{\text{dR}}^0(S^1) = \mathbb{R} \rightarrow \\ H_{\text{dR}}^1(\mathbb{RP}^2) \rightarrow H_{\text{dR}}^1(\mathbb{RP}^1) = \mathbb{R} \rightarrow H_{\text{dR}}^1(S^1) = \mathbb{R} \rightarrow H_{\text{dR}}^2(\mathbb{RP}^2) = 0. \end{aligned}$$

Since the alternating sum of dimension of an exact sequence is 0, $H_{\text{dR}}^1(\mathbb{RP}^2) = 0$.

Now we consider $n \geq 3$. If $1 \leq k \leq n-2$, the exact sequence shows that $H_{\text{dR}}^k(\mathbb{RP}^n) = H_{\text{dR}}^k(\mathbb{RP}^{n-1})$. It follows that $H_{\text{dR}}^k(\mathbb{RP}^n) = H_{\text{dR}}^k(\mathbb{RP}^{k+1})$ for $n \geq k+1 \geq 3$. The remaining exact sequence writes

$$\begin{aligned} H_{\text{dR}}^{n-2}(S^{n-1}) = 0 \rightarrow H_{\text{dR}}^{n-1}(\mathbb{RP}^n) \rightarrow H_{\text{dR}}^{n-1}(\mathbb{RP}^{n-1}) \\ \rightarrow H_{\text{dR}}^{n-1}(S^{n-1}) = \mathbb{R} \rightarrow H_{\text{dR}}^n(\mathbb{RP}^n) \rightarrow H_{\text{dR}}^n(\mathbb{RP}^{n-1}) = 0. \end{aligned}$$

Taking $n = 2m$, we have

$$0 \rightarrow H_{\text{dR}}^{2m-1}(\mathbb{RP}^{2m}) \rightarrow H_{\text{dR}}^{2m-1}(\mathbb{RP}^{2m-1}) = \mathbb{R} \rightarrow \mathbb{R} \rightarrow H_{\text{dR}}^{2m}(\mathbb{RP}^{2m}) = 0,$$

hence $H_{\text{dR}}^{2m-1}(\mathbb{RP}^{2m}) = 0$. Taking $n = 2m+1$, we have

$$0 \rightarrow H_{\text{dR}}^{2m}(\mathbb{RP}^{2m+1}) \rightarrow H_{\text{dR}}^{2m}(\mathbb{RP}^{2m}) = 0 \rightarrow \mathbb{R} \rightarrow H_{\text{dR}}^{2m+1}(\mathbb{RP}^{2m+1}) = \mathbb{R} \rightarrow 0,$$

hence $H_{\text{dR}}^{2m}(\mathbb{RP}^{2m+1}) = 0$. We see that

$$H_{\text{dR}}^k(\mathbb{RP}^n) = \begin{cases} \mathbb{R}, & k = 0 \text{ or } k = n \text{ is odd} \\ 0, & \text{else.} \end{cases}$$

□

Exercise 7.4. Let M be a compact oriented manifold. Prove that if $\dim M = 4n+2$, then its Euler characteristic $\chi(M)$ is even.

Solution. Let $b_i = \dim H_{\text{dR}}^i(M)$ be the i -th betti number of M . By Poincaré duality, $b_{4n+2-i} = b_i$, so the Euler characteristic writes

$$\chi(M) = \sum_{k=0}^{4n+2} (-1)^k b_k = b_{2n+1}.$$

Now consider the pairing defined by Poincaré duality:

$$A: H_{\text{dR}}^{2n+1}(M) \times H_{\text{dR}}^{2n+1}(M) \rightarrow \mathbb{R},$$

which is skew-symmetric and non-degenerate. $\det A = \det(-A) = (-1)^{b_{2n+1}} \det A$ is non-zero, hence $\chi(M) = b_{2n+1}$ is even. □

Exercise 7.5. Complete following two questions on mapping degree.

- (1) Let $f: \mathbb{T}^n \rightarrow \mathbb{T}^n$ be the map $f(e^{\sqrt{-1}\theta_1}, \dots, e^{\sqrt{-1}\theta_n}) = (e^{\sqrt{-1}k_1\theta_1}, \dots, e^{\sqrt{-1}k_n\theta_n})$.
- (2) Prove that there does not exist a map $S^2 \times S^2 \rightarrow \mathbb{CP}^2$ with odd degree.

Exercise 7.6. (1) Under local coordinates $\theta_1, \dots, \theta_n$, the map writes $(\theta_i) \mapsto (k_1\theta_i + 2l_i\pi)$, where $l_i \in \mathbb{Z}$ are constants. $\omega = \theta_1 \wedge \dots \wedge \theta_n$ is a well-defined volume form on $\mathbb{T}^n$, so

$$\int_{\mathbb{T}^n} f^*\omega = k_1 \cdots k_n \int_{\mathbb{T}^n} \omega.$$

By definition, we have $\deg f = k_1 \cdots k_n$.

(2) We need to recall some algebraic topology. The degree can be defined using integral cohomology, namely, $f^*\omega = \deg f \alpha$, where $\omega \in H^4(\mathbb{CP}^2; \mathbb{Z})$, $\alpha \in H^4(S^2 \times S^2; \mathbb{Z}) = H^2(S^2; \mathbb{Z}) \times H^2(S^2; \mathbb{Z})$. Moreover, this definition is compactible with degree in de Rham theory, in the sense that $H^*(M; \mathbb{Z}) \rightarrow H^*(M; \mathbb{R}) = H_{\text{dR}}^*(M)$ is a graded ring morphism and sends generators to generators.

Let β be the generator of $H^2(S^2; \mathbb{Z})$, then $\beta_1 \cup \beta_2$ is a generator of $H^4(S^2 \times S^2; \mathbb{Z}) = H^2(S^2; \mathbb{Z}) \times H^2(S^2; \mathbb{Z})$, where β_1, β_2 are (distinguished) copies of β . Let $\omega_{\text{FS}} \in H^2(\mathbb{CP}^2; \mathbb{Z})$ be the Fubini–Study form. Then ω_{FS}^2 is the generator of $H^4(\mathbb{CP}^2; \mathbb{Z})$. Write $f^*\omega_{\text{FS}} = b_1\beta_1 + b_2\beta_2$ with $b_i \in \mathbb{Z}$. Then we have

$$f^*\omega_{\text{FS}}^2 = (b_1\beta_1 + b_2\beta_2)^2 = 2b_1b_2\beta_1 \cup \beta_2,$$

which implies $\deg f = 2b_1b_2$ is even.

8 HOMEWORK FOR LECTURE 8

Exercise 8.1. Complete the following question on Hodge–Laplace operator.

- (i) Let M be a connected closed manifold and $f: M \rightarrow \mathbb{R}$ be a smooth function. Fix a volume form ω on M . Prove that $\Delta f = 0$ or $\Delta(f\omega) = 0$ if and only if f is a constant function.
- (ii) Under the same hypothesis of (1) above. Prove that $\int_M f\omega = 0$ if and only if there exists a smooth function $g: M \rightarrow \mathbb{R}$ such that $\Delta g = f$.

Solution. (i) First assume $\Delta f = 0$. Then $(f, \Delta f) = (df, df) = 0$, so $df = 0$ on M and f is a constant.

Now assume $\Delta f\omega = 0$. $(f\omega, \Delta f\omega) = (\delta(f\omega), \delta(f\omega)) = 0$. Since $\delta(f\omega) = (-1)^{n(n-1)+1} * d * (f\omega) = (-1)^{n(n-1)+1} * (df)$ and $*$ is an isomorphism, we have $df = 0$ and f is a constant function.

(ii) If $f = \Delta g$, $\int_M f\omega = (f, 1) = (g, \Delta 1) = 0$.

Conversely, assume $\int_M f\omega = 0$. By Hodge theorem,

$$\Omega^k(M) = \mathcal{H}^k(M) \oplus d(\Omega^{k-1}(M)) \oplus \delta(\Omega^{k+1}(M)).$$

$f = f_0 + \delta\alpha$ for some $f_0 \in \mathcal{H}^0(M)$ and $\alpha \in \Omega^1(M)$. Then f_0 must be a constant, and

$$0 = \int_M f\omega = f_0 \text{vol}(M) + (\delta\alpha, 1) = f_0 \text{vol}(M) + (\alpha, d1) = f_0 \text{vol}(M),$$

so $f_0 = 0$ and $f = \delta\alpha$. Again, we may write $\alpha = \alpha_0 + dg + \delta\beta$, for $\alpha_0 \in \mathcal{H}^1(M)$, $g \in C^\infty(M)$ and $\beta \in \Omega^2(M)$. Hence $f = \delta\alpha_0 + \delta dg + \delta^2\beta = \Delta g$. $\square$

Exercise 8.2. M is a smooth manifold with $\dim M = 3$. A contact 1-form on M is a form $\alpha \in \Omega^1(M)$ such that $d\alpha \wedge \alpha$ is a volume form. Complete the following questions.

(1) Prove that the hyperplane field $\mathcal{D}^2$ defined by

$$\mathcal{D}^2(p) = \ker \alpha(p)$$

for any $p \in M$ is not integrable anywhere (called completely non-integrable).

Such a completely non-integrable $\mathcal{D}^2$ is called a contact structure on M .

(2) Following the terminology in (1) right above, for $\mathbb{T}^3 = (\mathbb{R}/\mathbb{Z})^3$ in coordinate (x, y, z) , prove that $\mathcal{D}^2$ defined as follows,

$$\mathcal{D}^2 = \text{span} \left(\frac{\partial}{\partial z}, \cos(2\pi z) \frac{\partial}{\partial x} - \sin(2\pi z) \frac{\partial}{\partial y} \right)$$

is a contact structure on $\mathbb{T}^3$.

Solution. (1) Let $X, Y \in \mathcal{D}^2$. Then

$$d\alpha(X, Y) = Y\alpha(X) - X\alpha(Y) - \alpha([X, Y]).$$

Since $\alpha \wedge d\alpha$ is a volume form, $d\alpha$ is non-degenerate on every $\mathcal{D}^2(p)$. Therefore, we can always find $X, Y \in \mathcal{D}^2$ such that $d\alpha(X, Y) \neq 0$, hence $\alpha([X, Y]) \neq 0$ and $[X, Y] \notin \mathcal{D}^2(p)$.

(2) Let $\alpha = Adx + Bdy + Cdz$. We need to find A, B, C such that

$$\alpha\left(\frac{\partial}{\partial z}\right) = C = 0,$$

$$\alpha\left(\cos(2\pi z)\frac{\partial}{\partial x} - \sin(2\pi z)\frac{\partial}{\partial y}\right) = A\cos(2\pi z) - B\sin(2\pi z) = 0.$$

If we take $A = \sin(2\pi z)$ and $B = \cos(2\pi z)$, then α is a well-defined global 1-form on $\mathbb{T}^3$.

$$\begin{aligned} d\alpha \wedge \alpha &= (\sin(2\pi z)dy \wedge dz - \cos(2\pi z)dx \wedge dz) \wedge (\sin(2\pi z)dx + \cos(2\pi z)dy) \\ &= dx \wedge dy \wedge dz, \end{aligned}$$

so α is a contact structure on $\mathbb{T}^3$. □

Exercise 8.3. Use Sard's theorem and stereographic projection to prove that S^n is simply connected for $n \geq 2$.

Solution. Let $\gamma: S^1 \rightarrow S^n$ be a loop based on $p \in S^n$. Then γ is path homotopic to a smooth map, so we may assume γ is smooth. By Sard's theorem, $\gamma(S^1)$ has measure 0 in S^n , which implies γ is not surjective. By considering the stereographic projection with respect to a point not lying inside $\gamma(S^1)$, we have $\gamma: S^1 \rightarrow U \subset S^n$ where U is diffeomorphic to $\mathbb{R}^n$. Since $\mathbb{R}^n$ is a convex domain, γ retracts to the constant loop at p . □